UE26MA141A Engineering Mathematics – I · Unit 2 · Higher-Order Linear Differential Equations

Graphs and diagrams

Every graph and diagram in scope, redrawn as clean vector figures.

Class 006 · Particular integral Type 4 ($e^{ax}V$) and Type 5 (xV) · compiled 16 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
added for clarity	Standard context that is not in the class material
transcript only	Said in the lecture but not on any slide
video only	Seen only in a frame of the class video
not on current slides	From older material or past papers, not this year's slides
S006·5	Unit 2 deck 006 (“Particular integral of Type 4”), slide 5 (rendered .pptx)
T006 00:04:12	Unit 2 AV-summary transcript “Particular integral of standard functions – Type 4” (17 min), segment start
T006b 00:12:47	Unit 2 AV-summary transcript “Particular integral – Type 3 and Type 4” (bucket 006, 15.5 min); only its Type 4 part (00:09:55–00:15:25) is cited here. The class 005 pack cites the same file's Type 3 part as T006a.
CW U2 P18	Unit 2 class-work problem 18
ESA Jan-24 s3 2c	Past paper: UE23MA141A Jan 2024 set 3, question 2.c

Contents

1. Figure G1.1 · The PI $\frac{e^x \cos x}{2}$ oscillates inside its envelope $\pm\frac{e^x}{2}$
2. Figure G2.1 · Strip the exponential: $y_p e^{-3x}$ is the Type 3 polynomial
3. Figure G3.1 · Type 5: the PI of $(D^2-1)y = x\sin 3x$ grows like $\frac{x}{10}$

The deck has no graphs, so every figure is added for clarity. Colours follow the Okabe–Ito order: blue = the particular integral, vermilion = a second curve (the envelope $\pm e^{ax}$ or $\pm x$, or general solutions), green = a third (the shifted, exponential-free part); grey for guides. Every plotted PI was checked by substituting it back with sympy.

1 The Type 4 rule: $X = e^{ax}V$ (exponential shift)

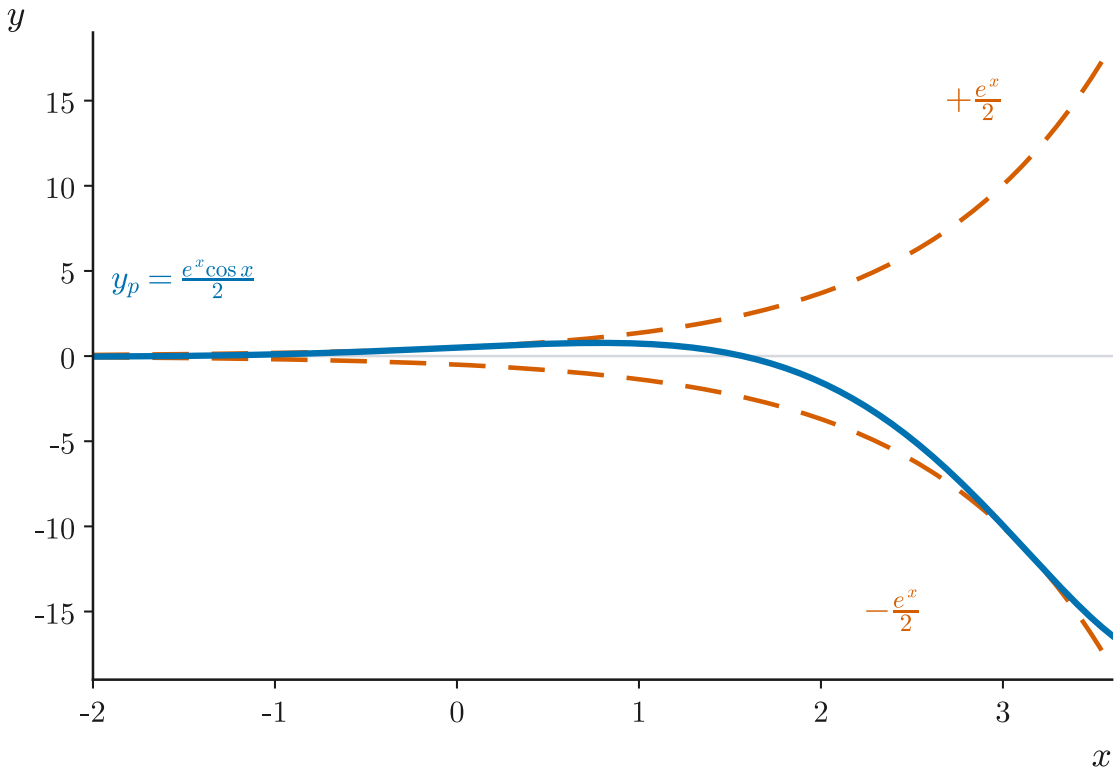


Figure G1.1. The PI $\frac{e^x \cos x}{2}$ oscillates inside its envelope $\pm\frac{e^x}{2}$. ESA ★★

ADDED FOR CLARITY S006·5 · S006·6

How to read it. The PI of $(D^2 - 2D + 4)y = e^x \cos x$ (blue) is the shifted problem's answer $\frac{\cos x}{2}$ multiplied by e^x: it touches the envelope $\pm\frac{e^x}{2}$ (vermilion dashed) where $\cos x = \pm 1$ and crosses zero where $\cos x = 0$. The factor e^{ax} survives the shift unchanged; only the oscillating part V is operated on.

2 Type 4 with a polynomial: $x^m e^{ax}$ and mixed right-hand sides

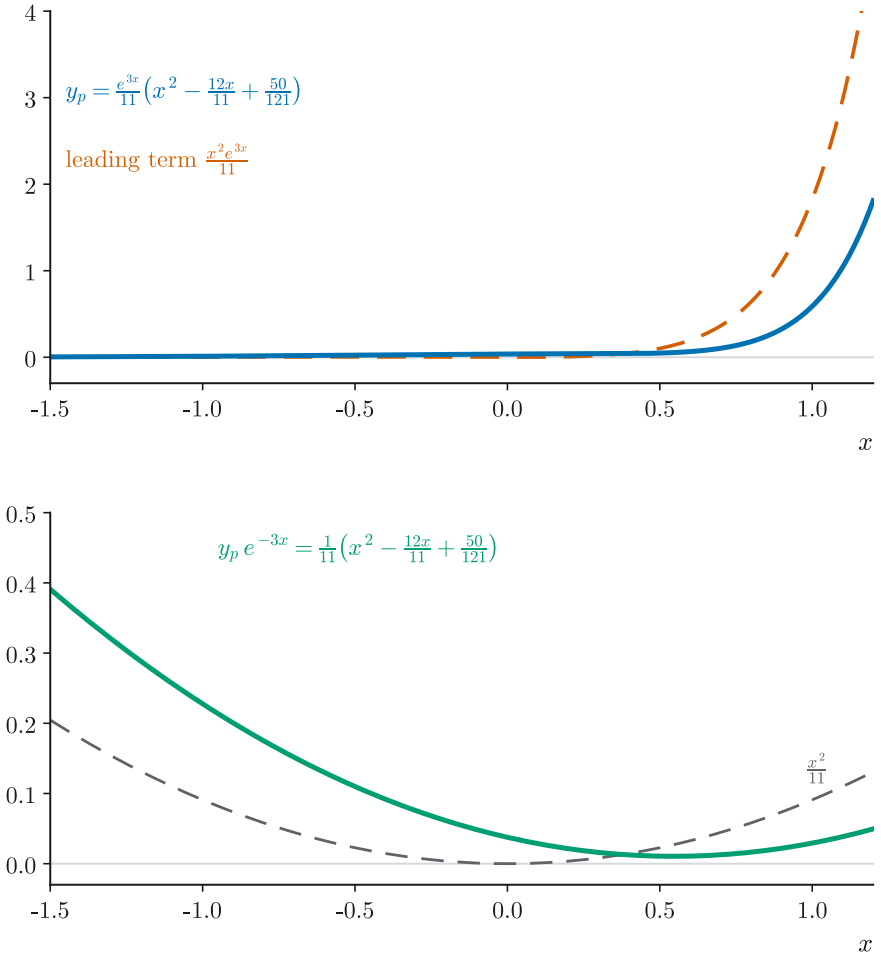


Figure G2.1. Strip the exponential: $y_p e^{-3x}$ is the Type 3 polynomial. ESA ★★

ADDED FOR CLARITY S006·11 · S006·12

How to read it. Top: the PI $y_p = \frac{e^{3x}}{11}\left(x^2 - \frac{12x}{11} + \frac{50}{121}\right)$ of $(D^2+2)y = x^2 e^{3x}$ (blue) next to $\frac{x^2 e^{3x}}{11}$ (vermilion dashed), the leading term alone. Bottom: divided by e^{3x}, the PI is the polynomial $\frac{1}{11}\left(x^2 - \frac{12x}{11} + \frac{50}{121}\right)$ (green), the answer of the shifted Type 3 problem $\frac{1}{D^2+6D+11}x^2$; the correction terms $-\frac{12x}{121} + \frac{50}{1331}$ move its vertex off the origin.

3 The Type 5 rule: $X = xV$

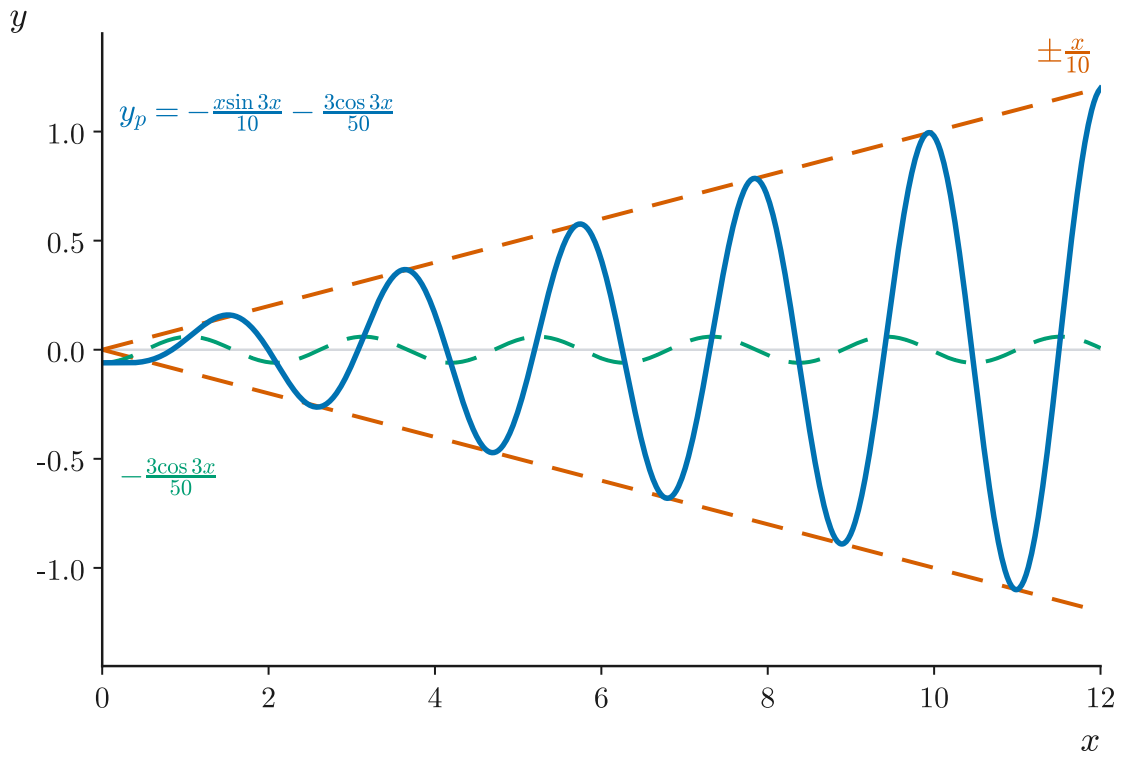


Figure G3.1. Type 5: the PI of $(D^2-1)y = x\sin 3x$ grows like $\frac{x}{10}$. ESA ★★

ADDED FOR CLARITY CW U2 P19

How to read it. The PI $-\frac{x\sin 3x}{10} - \frac{3\cos 3x}{50}$ (blue) has an amplitude that grows linearly, like the right-hand side $x\sin 3x$ scaled by $\frac{1}{f(-9)} = -\frac{1}{10}$; the lines $\pm\frac{x}{10}$ (vermilion dashed) are the envelope of the first term. The correction $-\frac{3\cos 3x}{50}$ (green dashed) is the $\frac{f'(D)}{(f(D))^2}V$ term: small, of fixed amplitude, and it dominates only near $x = 0$.